

RH-GHM-NCHG

C2 Ledger

Current checkpoint: transverse exactness, ray superconnection, determinant-line correction and global character target

This ledger freezes the proof architecture after the latest C2 run. It records what is now exact, what was corrected, what remains only locally supported, and the single global localization/character identity that is still open. It is not an assertion that the Riemann Hypothesis has been proved.

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1 Purpose and epistemic freeze

The previous ledger and Supplemental I left C2 in a deliberately broad form: a global operator family, compatibility with the differential, cancellation of exact/coexact sectors, and a determinant identity whose nontrivial divisor is the completed zeta function. Supplemental I also separated C1 existence from C1 faithfulness. The latest calculation changes the geometry of that open problem in three important ways.

Checkpoint update.

1. The transverse physical-sector constraint is no longer an axiom at the finite ray level: the transverse scalar and Cartan generator are explicitly Q -exact.
2. The same Cartan generator is also an explicit history boundary, giving a genuine low-arity chain intertwiner rather than a type analogy.
3. The old C2-X slogan “exact/coexact contribution equals one” is too crude for determinant lines. Exact states disappear from cohomology, but the acyclic sector can carry nontrivial analytic torsion. Local Euler factors live naturally in this torsion data.

The resulting final boss is therefore sharpened from “prove positivity” or “make zeros physical” to a global, zero-blind character/localization identity between an arithmetic scaling representation and the localized determinant line of the GHM time-ray superconnection.

1.1 Status language

Tag	Meaning
VERIFIED	direct finite-dimensional or symbolic derivation completed; no RH input.
SOURCE-EXACT	already explicitly present in NCHG/QMS source material.
SUPPORTED /	structurally matched to standard external machinery, but full
AUDIT	normalization/global theorem not reconstructed here.
PARTIAL / ACTIVE	a real map or local theorem exists, but not yet all places/arities/completions.
CORRECTED	an earlier formulation was too strong or in the wrong category and has been replaced.
OPEN BOSS	unresolved theorem carrying actual global proof burden.

2 Frozen inputs inherited from the main ledger

The following inputs remain unchanged.

2.1 Seed algebra and derived plane

The nonassociative real algebra is

$$A = \text{span}_{\mathbb{R}}\{1, h, q\}, \quad h^2 = q, \quad q^2 = q, \quad hq = 1, \quad qh = -1.$$

Its commutator quotient $V = A/\mathbb{R}1$ carries the symplectic form

$$\omega((b, c), (e, f)) = bf - ce.$$

The trace-free associator image is $\mathfrak{sl}_2(\mathbb{R})$ with the canonical split-Clifford triple

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

so that

$$H^2 = S^2 = I, \quad J^2 = -I, \quad HS = J = -SH.$$

The positive metric on V is

$$e(u, v) = -\omega(u, Jv).$$

The main ledger independently verified that the associator map $V \otimes V \rightarrow \text{End}(V)$ has determinant 3 in the natural basis, hence is an isomorphism in characteristic zero.

2.2 History/cohomology backbone

The NCHG history complex is

$$\mathcal{H}_n = C_\bullet(K_n) \otimes A^{\otimes n}, \quad \mathcal{H} = \bigoplus_{n \geq 1} \mathcal{H}_n,$$

with cellular differential $d^2 = 0$. At arity three, the oriented reassociation edge α obeys

$$d\alpha(x, y, z) = (xy)z - x(yz) = [x, y, z].$$

The framework already contains homotopy-transfer data and a BRST/BV physical quotient of the form

$$\mathcal{H}_{\text{phys}} = \ker Q_{\text{BRST}} / \text{im } Q_{\text{BRST}},$$

while the full cyclic nondegeneracy and quantum anomaly cancellation remain separate proof obligations.

2.3 Half-form and centered spectral coordinate

The spinor line is the half-form

$$\mathcal{L} \simeq K^{1/2},$$

and all spectral geometry is expressed in the centered variable

$$\lambda = s - \frac{1}{2} = \delta + i\gamma.$$

The already-verified finite time-ray Hodge relation is

$$\Delta_\delta = \delta^2 I.$$

3 The normalized time-ray complex

Fix an oriented unit ray $r \in V$ and define the quarter-turned normal

$$n := -Jr.$$

Then (r, n) is orthonormal for the positive metric e and $\omega(r, n) = 1$.

Define the graded 1-2-1 complex

$$C_r^{-1} = \mathbb{C}e_-, \quad C_r^0 = \mathbb{C}r \oplus \mathbb{C}n, \quad C_r^1 = \mathbb{C}e_+.$$

The differential is

$$q_r e_- = r, \quad q_r r = 0, \quad q_r n = e_+, \quad q_r e_+ = 0.$$

In the ordered basis (e_-, r, n, e_+) ,

$$q_r = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad q_r^2 = 0.$$

C2-T0 – normalized ray differential: VERIFIED. Direct multiplication gives $q_r^2 = 0$.

4 Transverse exactness is explicit

4.1 Scalar contraction

Define a degree -1 homotopy h_r by

$$h_r r = e_-, \quad h_r e_+ = n, \quad h_r e_- = h_r n = 0.$$

Its matrix is

$$h_r = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Exact multiplication yields

$$q_r h_r + h_r q_r = I.$$

For the centered spectral differential

$$Q_\delta := \delta q_r,$$

we obtain

$$\{Q_\delta, h_r\} = \delta I. \quad (4.1)$$

C2-T1 – scalar transverse exactness: VERIFIED. The real growth parameter enters as a literal BRST/Hodge-exact operator.

This is stronger than the previous statement $\Delta_\delta = \delta^2 I$: the transverse scalar itself has an explicit contracting primitive.

4.2 Cartan transverse generator

Define a second degree -1 map b_r by

$$b_r r = e_-, \quad b_r e_+ = -n, \quad b_r e_- = b_r n = 0.$$

Then

$$\{q_r, b_r\} = \text{diag}(1, 1, -1, -1).$$

Restricting to the middle ray plane $V = \text{span}\{r, n\}$,

$$\{q_r, b_r\}|_V = H_r = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (4.2)$$

Therefore

$$\{Q_\delta, b_r\}|_V = \delta H_r. \quad (4.3)$$

C2-T2 – Cartan transverse exactness: VERIFIED. The noncompact H direction is not merely damped by a localization semigroup; it is Q -exact in the normalized ray complex.

4.3 Operator-valued lift

Let A now denote a densely-defined self-adjoint operator on an auxiliary Hilbert carrier $\mathcal{K}$. Set

$$Q_A = q_r \widehat{\otimes} A.$$

Since $q_r^2 = 0$,

$$Q_A^2 = 0.$$

With the normalized adjoint on the finite ray factor,

$$q_r q_r^\dagger + q_r^\dagger q_r = I,$$

so

$$\Delta_A = Q_A Q_A^\dagger + Q_A^\dagger Q_A = I \widehat{\otimes} A^2. \quad (4.4)$$

Moreover

$$\{Q_A, h_r \otimes I\} = I \otimes A, \quad (4.5)$$

and on the middle plane

$$\{Q_A, b_r \otimes I\}|_V = H_r \otimes A. \quad (4.6)$$

Thus an A -eigenmode of eigenvalue δ has Hodge cost δ^2 .

C2-T3 – operator-valued Hodge lift: VERIFIED. No zeta function, zero location, or RH assumption is used.

5 The same H is a history boundary

On V the associator operators are

$$A_{hh} = \begin{pmatrix} 0 & -1 \\ 0 & 1 \end{pmatrix}, \quad A_{hq} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad A_{qh} = \begin{pmatrix} -1 & 0 \\ -1 & -2 \end{pmatrix}, \quad A_{qq} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Direct summation gives

$$A_{hq} + A_{qh} + A_{qq} = H. \quad (5.1)$$

Define the operator-valued history edge

$$\mathbf{b}_H(w) := \alpha(h, q, w) + \alpha(q, h, w) + \alpha(q, q, w).$$

Using the source-exact identity $d\alpha = [\ , \]$,

$$\begin{aligned} d_{\text{hist}} \mathbf{b}_H(w) &= [h, q, w] + [q, h, w] + [q, q, w] \\ &= Hw. \end{aligned}$$

Hence

$$d_{\text{hist}} \mathbf{b}_H = H. \quad (5.2)$$

Combining (4.2) and (5.2),

$$H = d_{\text{hist}} \mathbf{b}_H = \{q_r, b_r\}|_V. \quad (5.3)$$

C2-T4 – transverse history/ray intertwiner: VERIFIED at low arity. This closes the transverse generator slot of C1-I as an exact commutative square, not merely a type match.

A convenient square is

$$\begin{array}{ccc} \mathbf{b}_H & \xrightarrow{d_{\text{hist}}} & H \\ \downarrow \mathcal{R} & & \downarrow \mathcal{R} \\ b_r & \xrightarrow{\{q_r, \cdot\}} & H_r \end{array}$$

with $\mathcal{R}(\mathbf{b}_H) = b_r$ and $\mathcal{R}(H) = H_r$.

6 Vertical exactness versus horizontal phase

The next question is why H is removable while J is not. This is now a category statement rather than a positivity slogan.

6.1 Exact degree-zero endomorphisms preserve the ray line

Let B be an arbitrary degree -1 map. Write

$$B(r) = ae_-, \quad B(n) = be_-, \quad B(e_+) = cr + dn.$$

Then on V ,

$$\{q_r, B\}|_V = \begin{pmatrix} a & b+c \\ 0 & d \end{pmatrix}. \quad (6.1)$$

Thus vertical exact endomorphisms form the upper-triangular/Borel sector that preserves the ray line $\mathbb{C}r$.

But

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

does not preserve $\mathbb{C}r$: it sends r to a transverse ray. Therefore J is not another vertical exact gauge generator in the fixed-ray complex.

C2-J0 – vertical/horizontal category split: VERIFIED. H lies in the exact vertical sector; J moves between ray fibers.

6.2 Ray bundle and horizontal J connection

Let

$$r_\theta = e^{-\theta J} r_0.$$

The family of ray differentials is related by

$$q_\theta = G_\theta q_0 G_\theta^{-1}, \quad G_\theta = \text{diag}(1, e^{-\theta J}, 1).$$

Define

$$\mathcal{J} = \text{diag}(0, J, 0), \quad \nabla_\theta = \partial_\theta + \mathcal{J}.$$

A direct derivative gives

$$[\nabla_\theta, q_\theta] = 0. \quad (6.2)$$

Hence J defines a horizontal connection on the bundle of ray complexes.

Its holonomy is

$$\text{Hol}_{0 \rightarrow \theta}(\nabla) = e^{-\theta J}.$$

For a full oriented spinor circle,

$$e^{-2\pi J} = I,$$

while a projective half-turn gives

$$e^{-\pi J} = -I. \quad (6.3)$$

This is the finite-dimensional origin of the spinor double cover and is compatible with the half-form sign already frozen in the main ledger.

C2-J1 – horizontal phase connection: VERIFIED. The compact J direction is geometrically horizontal rather than a surviving vertical ghost.

7 A genuine nonseparable ray superconnection

Define the superconnection on the ray-circle family

$$\mathbb{A}_\delta = Q_\delta + d\theta \nabla_\theta, \quad (7.1)$$

with $Q_\delta = \delta q_\theta$. Since $Q_\delta^2 = 0$, $[\nabla_\theta, Q_\delta] = 0$, and $d\theta \wedge d\theta = 0$,

$$\mathbb{A}_\delta^2 = 0. \quad (7.2)$$

C2-S0 – local/family superconnection: VERIFIED. This is the first explicit nonseparable object that simultaneously carries vertical BRST exactness and horizontal phase holonomy.

The decomposition is now structural:

$$\text{vertical/radial} = Q\text{-exact}, \quad \text{horizontal/angular} = J\text{-phase}. \quad (7.3)$$

This mirrors the older QMS chronology split in which the ordering coordinate is gauge-like while the reactive coordinate is an internal phase/history direction.

8 Local arithmetic transfer and chiral determinant

For a prime p put

$$\ell_p = \log p, \quad \theta_p = \gamma \ell_p.$$

The local transfer operator is written

$$U_p(s) = p^{-1/2} e^{-\delta \ell_p} e^{-\gamma \ell_p J}, \quad s = \frac{1}{2} + \delta + i\gamma. \quad (8.1)$$

This separates three geometrically distinct factors:

$$\text{half-density} \times \text{radial amplitude} \times J\text{-holonomy}.$$

The radial generator is exact by (4.1), while the angular factor is horizontal by (6.2).

Complexify the ray plane and let

$$V^{1,0} = \ker(J - iI), \quad V^{0,1} = \ker(J + iI).$$

On $V^{1,0}$,

$$U_p(s) = p^{-1/2-\delta} e^{-i\gamma \log p} = p^{-s}.$$

Therefore

$$\det_{V^{1,0}}(I - U_p(s)) = 1 - p^{-s}. \quad (8.2)$$

On the conjugate chirality,

$$\det_{V^{0,1}}(I - U_p(s)) = 1 - p^{-\bar{s}}.$$

Consequently the real doubled determinant is

$$\det_{\mathbb{R}}(I - U_p(s)) = |1 - p^{-s}|^2. \quad (8.3)$$

C2-L0 – local chiral Euler factor: VERIFIED. The local holomorphic Euler factor is the determinant section of one J chirality; the paired real determinant is its Hermitian norm square.

8.1 Local Hodge/conductor response

The earlier local shift model

$$Q_{p,\delta} = I - p^{-1/2} e^{-\delta \log p} S$$

has Fourier symbol

$$\widehat{\Delta}_{p,\delta}(\theta) = |1 - p^{-1/2} e^{-\delta \log p} e^{i\theta}|^2.$$

Differentiating $-\log \widehat{\Delta}$ transversely at $\delta = 0$ produces the Poisson-kernel coefficient pattern matched in the main ledger to the local conductor operator formalism.

C2-L1 – local conductor identification: SUPPORTED / AUDIT. The structure and matrix coefficients were matched, but publication-grade Haar/Fourier normalization and constant-term audit remain to be frozen.

9 Critical correction: exact states can leave determinant torsion

Supplemental I stated C2-X in the form

$$\text{sdet}_{\text{reg}}(\mathcal{D}_A|_{\text{im } Q \oplus \text{im } Q^*}) = 1.$$

That formulation is now too strong.

An acyclic complex contributes no cohomology, but its determinant line can carry nontrivial analytic torsion. In the local arithmetic model, $I - p^{-s} S$ is invertible in its Euler region, yet its determinant section is precisely the local Euler factor. Thus the correct statement is not “exact contribution equals one” but rather:

Revised C2-X*. Exact/coexact states do not contribute physical cohomology classes. Their determinant-line contribution is encoded by a canonical acyclic torsion section. This torsion must be tracked, normalized and glued globally; it cannot be silently replaced by 1.

Symbolically, over a domain where the acyclic/Hodge splitting is defined,

$$\text{Det}(\mathcal{D}_A) \simeq \text{Det}(\mathcal{D}_A|_{H_Q}) \otimes \tau_{\text{ac}}(s), \quad (9.1)$$

where τ_{ac} is the analytic torsion section of the exact sector.

C2-X – CORRECTED. State cancellation survives; determinant triviality does not. Local Euler data is naturally torsion data.

This correction also clarifies the holomorphic/Hermitian split:

$$\xi(s) \text{ should appear as a holomorphic determinant section,}$$

while

$$|\xi(s)|^2 \text{ is the natural Hodge/Quillen-type norm square.}$$

The local formula (8.2)–(8.3) realizes this distinction literally.

10 What the global C2 theorem must now say

The local pieces are no longer the conceptual bottleneck. The global theorem must simultaneously do five things.

10.1 C2-G1: restricted-product superconnection

Construct an adelic/restricted-product complex $\mathcal{K}_A$ with a global superconnection $\mathbb{A}_A$ whose localizations recover:

- the normalized ray differential and its exact transverse generator;
- the J -horizontal connection;
- odd-prime Euler/conductor blocks;
- the dyadic metaplectic refinement;
- the archimedean Gaussian/half-form block.

OPEN BOSS

10.2 C2-G2: quotient and global character

The global object cannot be the direct sum of independent prime blocks. The main ledger already established the local-kernel no-go. One must take the arithmetic quotient and show that the resulting scaling action has a well-defined distributional/regularized character.

OPEN BOSS

10.3 C2-G3: torsion localization formula

Prove a global localization identity of the schematic form

$$\boxed{\chi_{\text{ar}} = \chi_{\text{GHM}}^{\text{loc}}} \quad (10.1)$$

where the right-hand side is computed from the physical/Hodge part together with the canonically normalized torsion of the exact sector.

This is the sharpened version of the former determinant identity. It is zero-blind: neither side may use nontrivial zero ordinates in its definition.

OPEN BOSS

10.4 C2-G4: determinant/divisor realization

A determinant-line form of (10.1) should produce

$$\boxed{\text{sdet}_{\text{reg}}(s - \Theta_A) = E(s) \xi(s), \quad E(s) \neq 0} \quad (10.2)$$

on the nontrivial-zero domain, with multiplicities preserved. The exact sector is not erased; its torsion is part of the normalization and local-to-global determinant construction.

OPEN BOSS

10.5 C2-G5: physical generator after localization

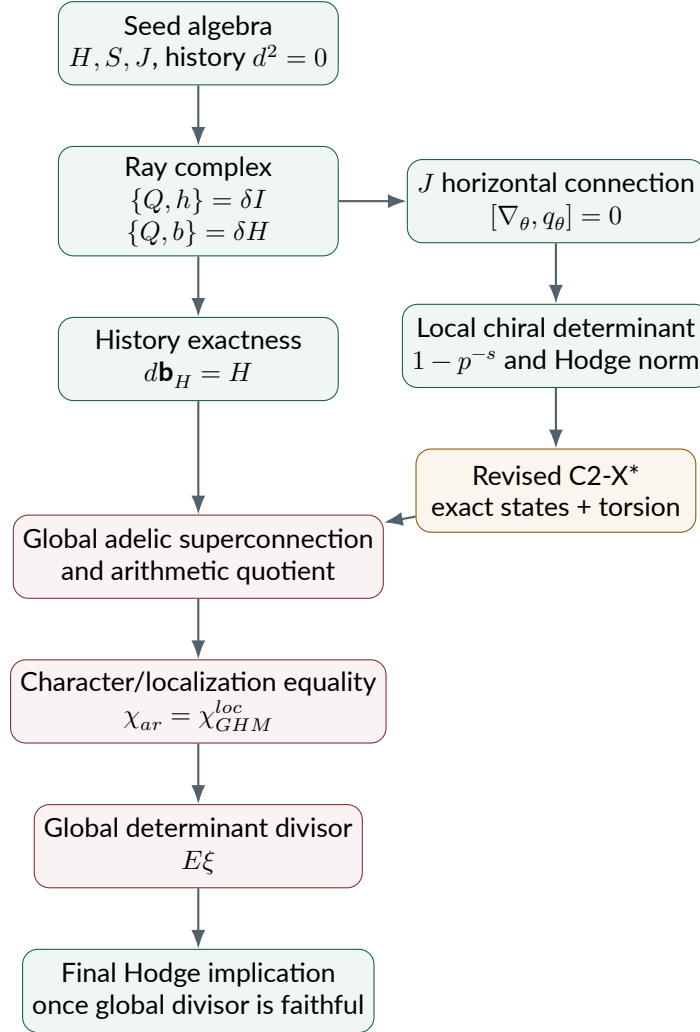
After passing to physical cohomology, the transverse generator must disappear by the already-verified exactness relation. The expected reduced generator is

$$\boxed{[\Theta_A] = \frac{1}{2}I + JP, \quad P = P^*,} \quad (10.3)$$

with $J^* = -J$ on the positive physical metric. This is no longer a free physical-sector postulate at the finite ray level; only the global identification of the arithmetic generator with this localized action remains open.

PARTIAL / ACTIVE

11 Updated dependency graph



12 Claim ledger for the C2 checkpoint

ID	Layer	Status	Claim / audit note
C2-T0	Ray differential	VERIFIED	$q_r^2 = 0$ on the normalized 1-2-1 complex.
C2-T1	Transverse scalar	VERIFIED	$\{Q_\delta, h_r\} = \delta I$. The radial growth scalar is Q -exact.
C2-T2	Cartan transverse	VERIFIED	$\{Q_\delta, b_r\} _V = \delta H_r$.
C2-T3	Operator lift	VERIFIED	$\Delta_A = I \otimes A^2$ and $I \otimes A, H \otimes A$ are exact by explicit homotopies.
C2-T4	History bridge	VERIFIED	$A_{hq} + A_{qh} + A_{qq} = H$ and $d_{hist} \mathbf{b}_H = H$.

ID	Layer	Status	Claim / audit note
C2-I0	C1-I transverse slot	VERIFIED	The low-arity history-to-ray square for H commutes exactly. This does not yet prove a global all-place functor.
C2-J0	Vertical exact algebra	VERIFIED	Exact degree-zero maps are upper triangular and preserve the ray line.
C2-J1	Horizontal phase	VERIFIED	J acts as a horizontal connection on the ray bundle; $[\nabla_\theta, q_\theta] = 0$.
C2-J2	Spinor holonomy	VERIFIED	$e^{-2\pi J} = I$ while the projective half-turn gives $e^{-\pi J} = -I$.
C2-S0	Superconnection	VERIFIED	$\mathbb{A}_\delta = Q_\delta + d\theta\nabla_\theta$ satisfies $\mathbb{A}_\delta^2 = 0$.
C2-L0	Local determinant	VERIFIED	One J chirality gives $1 - p^{-s}$; the real doubled determinant gives $ 1 - p^{-s} ^2$.
C2-L1	Local conductor	SUPPORTED / AUDIT	Transverse derivative reproduces the local Poisson-kernel/conductor structure; full normalization audit still needed.
C2-X0	Old exact cancellation	CORRECTED	"Acyclic superdeterminant = 1" is not generally correct as a determinant-line statement.
C2-X1	Revised exact sector	PARTIAL / ACTIVE	Exact/coexact states vanish from cohomology but carry canonical analytic torsion. A global torsion normalization theorem is still open.
C2-G1	Adelic glue	OPEN BOSS	Build the restricted-product superconnection including odd p , $p = 2$ and ∞ .
C2-G2	Arithmetic quotient	OPEN BOSS	Show the global quotient produces collective resonances/character rather than a direct-sum local kernel picture.
C2-G3	Character equality	OPEN BOSS	Prove zero-blind $\chi_{ar} = \chi_{GHM}^{loc}$. This is the current sharp boss theorem.
C2-G4	Determinant divisor	OPEN BOSS	Derive $\text{sdet}_{\text{reg}}(s - \Theta_A) = E(s)\xi(s)$ with E nowhere zero on the nontrivial-zero domain and multiplicities preserved.
C2-G5	Reduced generator	PARTIAL / ACTIVE	Finite transverse exactness is proved; global arithmetic identification with $\frac{1}{2} + JP$ remains part of G3/G4.
C2-F	Final implication	VERIFIED	Once the global divisor is faithfully localized to physical cohomology, the existing Hodge theorem forces zero transverse parameter.

13 False-positive and false-negative guards

13.1 False-positive prosecutor

The following moves are explicitly forbidden:

- defining the global operator, P , or the map $\mathcal{R}$ from known zero ordinates;
- replacing the arithmetic quotient by an independent direct sum of local Euler blocks;
- reading local invertibility or local positivity as global spectral faithfulness;
- asserting acyclic determinant contribution is 1 without a determinant-line/torsion proof;
- assuming that analytic continuation of a product preserves the same Hilbert kernel;
- treating the low-arity exact square $d\mathbf{b}_H = H \leftrightarrow \{q, b\} = H$ as an all-arity adelic theorem;
- inserting the dyadic or archimedean factors by hand after the fact.

13.2 False-negative defender

The following failures do not automatically kill the route:

- one particular completion may erase zero modes while a distributional/Frechet or determinant-line realization survives;
- local blocks may remain acyclic even when the global quotient carries collective resonances;
- strict equality may be replaced by a chain-homotopy equivalence, provided the induced torsion anomaly is tracked;
- a naive positive L^2 quotient can be trivial without implying the global character localization is impossible;
- a failed norm-square route does not automatically kill a holomorphic determinant-line route.

14 Minimal theorem package from this checkpoint

The C2 program can now be stated with fewer ad hoc ingredients.

C2 Closure Package (revised). Construct, without using nontrivial zero locations:

1. a global adelic/history superconnection localizing to the verified ray complex and J connection;
2. a determinant line with canonical torsion for the acyclic exact sector;
3. an arithmetic quotient and regularized character χ_{ar} ;
4. a localization theorem $\chi_{ar} = \chi_{GHM}^{loc}$;
5. the induced determinant-divisor identity $\text{sdet}_{\text{reg}}(s - \Theta_A) = E(s)\xi(s)$ with controlled E and multiplicities.

The finite transverse exactness and horizontal phase split are no longer assumptions inside this package.

This is materially sharper than Supplemental I. The unknown object is not “a positive physical sector” and not “a self-adjoint operator somewhere.” It is the global arithmetic character of a specific superconnection geometry whose local vertical, horizontal and determinant-line pieces are already fixed.

15 Independent symbolic verification appendix

For this ledger, the newly claimed finite identities were independently rechecked symbolically.

15.1 Ray matrices

With

$$q_r = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad h_r = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$b_r = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

exact multiplication gives

$$\begin{aligned} q_r^2 &= 0, & q_r h_r + h_r q_r &= I_4, \\ q_r b_r + b_r q_r &= \text{diag}(1, 1, -1, -1). \end{aligned}$$

15.2 Associator sum

The independently reconstructed associator matrices satisfy

$$A_{hq} + A_{qh} + A_{qq} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = H.$$

This matches the history-boundary derivation in Section 5.

16 Source anchors and references

Project-internal sources

1. *RH-GHM-NCHG Research Ledger*, especially the time-ray, Hodge, local conductor, false-positive/false-negative, and current dependency-graph sections.
2. *RH-GHM-NCHG Supplemental I: C1 Update, C1 Faithfulness and C2 Global Factorization Specification*, especially the C1-F hierarchy and original C2-Q/X/D/H contract.
3. *NCHG v0.2*, Appendix C (associahedral history resolution), Sections 6.3–6.8 (BV/BRST and chronology fields), and chronology closure sections.
4. *QMS Theory v1.0*, Chapter 7: chronology algebra $\mathbb{C}_J \oplus \mathbb{R}$, ordering coordinate, reactive phase, and $[D_t, D_u] = JD_\chi + \mathcal{F}_{tu}^\perp$.
5. *Associator Reactive Chern Pump*, for the exact CAR/Pauli/ $J^2 = -1$ chain and the positive regular-action metric.

External structural precedents used only as orientation

1. J.-F. Burnol, works on the local conductor operator and the explicit formula.
2. R. Meyer, work realizing zeta zeros through a scaling representation/character formula.
3. C. Deninger, regularized-determinant and dynamical-zeta program.
4. J.-M. Bismut and D. Freed, determinant-line and Quillen-type geometry for families of elliptic operators.

No external precedent above is treated as a proof of C2-G1–G4. Their role is to show that the requested global object has a known mathematical type; the actual zero-blind identification remains the open theorem.

C2 Ledger freeze – 12 August 2026

Current sharp boss: global character/localization equality, not transverse positivity.